Automated Radiology Impression Generation Using a Fine-Tuned Medical Language Model

Multi-Agent Design, Fine-Tuning, and Evaluation Summary

Abstract

This report describes a medical NLP task focused on generating radiology impressions from free-text findings. We present a baseline model, a multi-agent scoring system, and a LoRA fine-tuning workflow. Quantitative results show improved n-gram overlap and recall-based similarity after fine-tuning, while qualitative analysis highlights where baseline outputs remain limited in differential diagnosis and management recommendations.

1. Multi-Agent Design and Baseline Performance

Task 1.1 – Baseline model. We used the pretrained `m42-health/Llama3-Med42-8B` model to generate impression lists from radiology findings. The baseline pipeline loads the findings, performs deterministic generation, and writes outputs to a CSV file with `case_id` and `generated_impression` fields.

Prompt strategy. The prompt is designed to instruct the model to generate a Python list of concise disease impressions using formal medical terminology without including extra explanations. By specifying the output format and providing examples, the prompt helps ensure that the model outputs are consistent with the ground-truth format and easily comparable for evaluation.

Task 1.2 – Multi-agent scoring. A second evaluation agent scores each generated impression across eight criteria on a 0–5 scale. Table 1 and Figure 2 summarize the average scores.

Workflow overview. Once the baseline model generates a list of disease impressions for each set of findings, a panel of evaluation agents independently scores each output on **scientific terminology**, **coherence**, **specific diagnosis**, **correctness**, **differential diagnosis**, **management recommendations**, **comprehensiveness**, and **harmlessness/bias** using a 0–5 scale. The scores are then aggregated across the test set to produce summary metrics. These metrics provide a detailed assessment of the baseline model's performance.

Figure 1: Multi-agent evaluation workflow. The baseline model generates disease impressions from radiology findings, and evaluation agents score each output on eight criteria (0–5). Scores are aggregated across the test set to produce summary metrics.

Criterion	Average score
Scientific terminology	4.15
Coherence	4.14
Specific diagnosis	3.71
Differential diagnosis	1.90
Management recommendations	1.34
Correctness	3.83
Comprehensiveness	3.69
Harmlessness / Bias	4.98

Table 1: Multi-agent evaluation of baseline impressions.

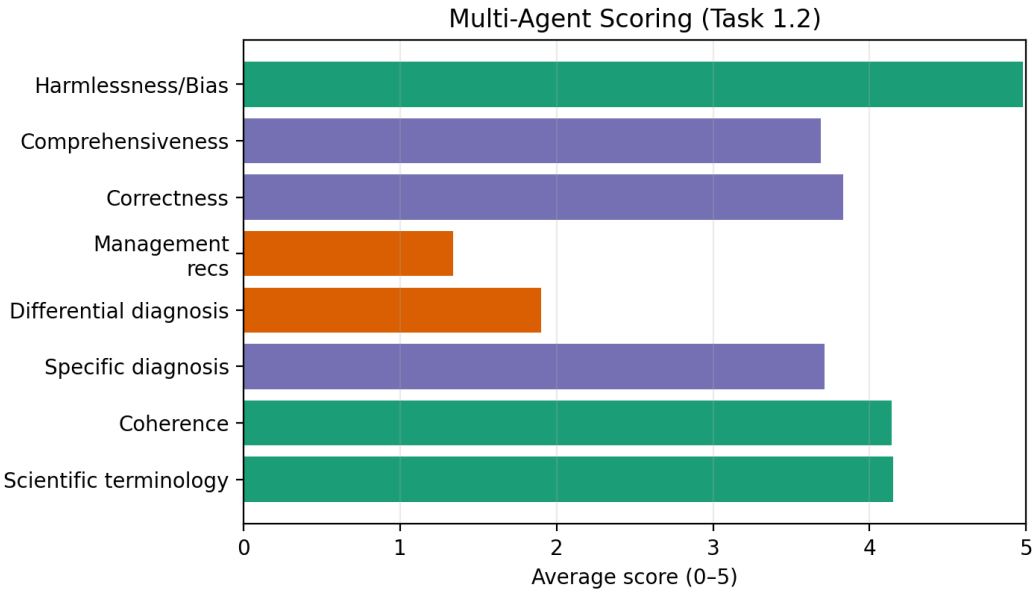

Figure 2: Baseline performance across eight evaluation criteria (multi-agent scoring on a 0–5 scale; averages across the evaluation set).

Interpretation. The baseline demonstrates strong terminology and safety, but weak differential diagnosis and management guidance, suggesting the model is competent at concise descriptions yet misses alternative explanations and follow-up recommendations.

2. Task 2 – Fine-Tuning Methodology

Baseline vs. fine-tuned. We compared the baseline model to a LoRA fine-tuned version. LoRA updates a small subset of attention weights, enabling efficient adaptation.

Training data and preprocessing. The fine-tuned model was trained on 1,200 radiology report findings, with 380 examples used for testing. No additional preprocessing was applied due to time constraints during model training. Potential preprocessing could include removing empty or

malformed entries and standardizing terminology, which could improve model consistency.

Training setup	Value
Epochs	7
Learning rate	1×10^{-5}
Batch size	6
LoRA rank r	8
LoRA α	16
Dropout	0.1
Target modules	`q_proj, k_proj, v_proj, o_proj`

Table 2: Fine-tuning configuration.

Evaluation metrics (380 test samples). Table 3 and Figure 3 show improvements across EM, BLEU, and ROUGE.

Model	EM	BLEU	ROUGE-1	ROUGE-2	ROUGE-L
Baseline	0.000	0.0456	0.1895	0.0900	0.1770
Fine-tuned	0.026	0.1420	0.4650	0.2590	0.4520

Table 3: Quantitative comparison of baseline and fine-tuned models.

Evaluation metrics. *Exact Match (EM)* is the proportion of predicted impressions that exactly match the reference. *BLEU* measures n-gram overlap between prediction and reference. *ROUGE-1*, *ROUGE-2*, and *ROUGE-L* are F1-scores based on unigram, bigram, and longest common subsequence overlap. Higher scores indicate closer alignment with the reference impressions.

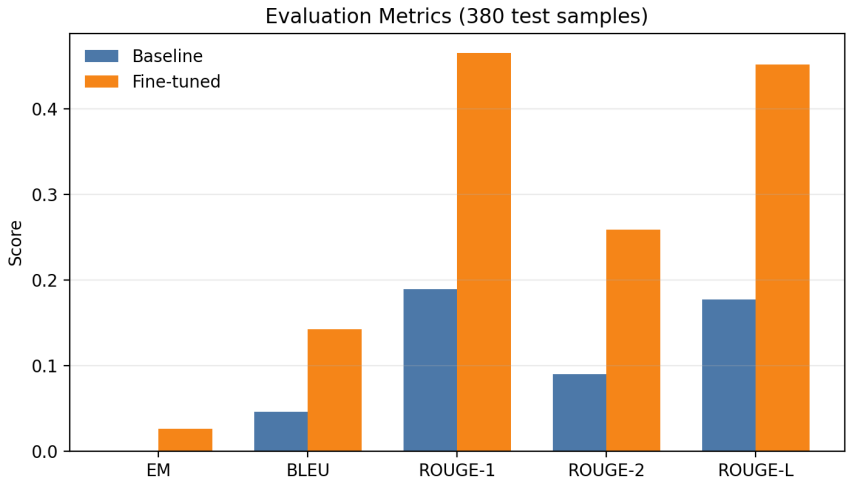

Figure 3: BLEU and ROUGE metrics for baseline vs. fine-tuned models across 380 test samples.

Qualitative comparison. The fine-tuned model produces longer, more complete impressions and includes occasional management suggestions. This aligns with the higher ROUGE metrics.

Quantitative improvement. ROUGE-L improved from 0.177 to 0.452, a relative increase of approximately 156%.

3. Example Outputs

Reference	Baseline Impression	Fine-Tuned Impression
Stone in the intravesical segment of the distal end of the right ureter, with right ureteral and renal hydronephrosis.	Normal liver parenchyma with no evidence of intrahepatic or extrahepatic bile duct dilation. Normal gallbladder, pancreas, spleen, and intestinal loops. No ascites.	No definite abnormalities on the whole abdomen CT scan.
Fatty liver, hepatic calcification., Multiple cysts in bilateral kidneys possible., Appendicitis	1. Multiple renal lesions, likely benign, need further evaluation. 2. Thickened intestinal wall and appendix, possible inflammatory changes or early malignancy. 3. Liver nodular lesion, need further evaluation. Recommendations: 1. Order abdominal CT with contrast to better evaluate the renal lesions and assess for any enhancement patterns. ...	Multiple cysts in both kidneys., Appendicitis suspected., Fatty liver.

Table 4: Example outputs illustrating concise baseline text and more comprehensive fine-tuned responses.

4. Failure Mode Analysis

During early experiments, the model occasionally produced degenerate outputs, such as "!!!!!" or unintelligible text. Investigation revealed that the root causes were not in the LoRA training loop itself, but in dataset and generation handling:

- **Improper label masking:** Prompt and padding tokens were not correctly masked, causing the model to learn from unintended tokens.
- **Invalid batches / NaN losses:** Some batches produced NaN gradients, leading to unstable learning.
- **Greedy decoding issues:** Initially, deterministic greedy decoding without repetition control caused repeated or nonsensical outputs.

Fixes implemented:

- Correctly masked prompt and padding tokens in the dataset.
- Skipped invalid batches to avoid NaN losses.
- Used deterministic greedy decoding with a mild repetition penalty (1.1).

- Removed invalid outputs and applied manual gradient clipping as needed.

After applying these changes, the model produced stable, coherent, and clinically plausible impressions. Early noisy or degenerate outputs disappeared, and the generated text aligned better with reference impressions.

5. AI-Assisted Workflow

AI was leveraged throughout the project to streamline coding, debugging, and reporting:

- **Coding assistance:** GitHub Copilot suggested code for data loading, tokenization, dataset definition, and building the training loop, saving time on boilerplate tasks and allowing focus on methodology.
- **Debugging:** AI tools (Codex, ChatGPT) helped diagnose why the model produced degenerate outputs and suggested practical fixes, improving efficiency in troubleshooting.
- **Report drafting and analysis:** AI supported summarizing methodology, explaining results, and comparing model outputs. Prism was used for data visualization and for generating tables and figures summarizing evaluation results.

Using AI in these ways accelerated development, reduced repetitive work, and allowed deeper focus on methodological decisions and result interpretation.

Reproducibility

All code for fine-tuning, inference, and evaluation scripts, as well as CSV outputs for baseline and fine-tuned models, are available at `https://github.com/absterk/MedDomain-LLM`. Pretrained and fine-tuned models can be downloaded following the instructions in the repository.

Summary

- The baseline model is strong in terminology and harmlessness but weak in differential diagnosis and management guidance.
- LoRA fine-tuning substantially improves BLEU and ROUGE, indicating closer alignment with reference impressions.
- ROUGE-L improved from 0.177 (baseline) to 0.452 (fine-tuned), representing a 156% increase and substantially closer alignment with reference impressions.
- Correct label masking and deterministic decoding eliminate degenerate outputs.
- AI assistance accelerates coding, debugging, and report composition.